

John P. Acosta

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EDUCATION

NEW JERSEY INSTITUTE OF TECHNOLOGY

Newark, NJ

Bachelor of Science in Computer Science, Ying Wu College of Computing

Expected May 2027

- **GPA:** 3.7 **Awards:** NSF S-STEM Scholar, Dean's List (All Semesters), Ron Andrews PGIM Scholar, Kupfrian Memorial Scholar

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Programming Language Concepts (C/C++), Operating Systems, Computer Networks, Artificial Intelligence, Data Science, Database Design & Management, Linear Algebra, Probability & Statistics

SKILLS

Languages: Python, Java, C/C++, JavaScript

Frameworks: FastAPI, React/Next.js

Databases: PostgreSQL, SQLite, Supabase

Data/ML: NumPy, pandas, scikit-learn, PyTorch, Transformers, PEFT

Cloud/DevOps: AWS, Azure, Terraform, Jenkins, GitHub Actions, CI/CD, IaC

Tools: Git/GitHub, Linux/Bash, Docker, LangChain, SLURM/HPC, pytest

PROFESSIONAL EXPERIENCE

Software Engineer Intern

Jun 2026 - Aug 2026

PGIM Technology DevOps

- Authored 7 reusable Terraform modules (Lambda, S3, KMS) migrating PGIM infrastructure off CloudFormation, cutting infrastructure code ~80% and average build time ~66% per application, as part of the shared IaC standards consolidating 7 siloed business units onto unified AWS and Azure infrastructure
- Built a FinOps cost-reporting dashboard (React, FastAPI, AWS S3) tracking \$1M+ in monthly cloud spend, replacing a 2 to 3 hour manual monthly report with self-serve visibility; backfilled 6 months of history via a Claude-based skill parsing archived Confluence PDFs
- Placed 2nd at PGIM's internal hackathon, leading 5 engineers to prototype DocDoctor, a documentation quality pipeline that automatically identifies stale Confluence pages, generates update suggestions, and notifies reviewers, improving the reliability of knowledge served to Copilot and Claude MCP at scale
- Launched a Power Platform training index (Dataverse, SharePoint, Canvas app) letting employees locate training resources through firm Copilot and subscribe to training session notifications across 7 business units

Applied AI Research Engineer Intern

Jan 2026 - Jun 2026

NJIT AIM Lab

- Designed a quality assurance framework across a PyQt desktop application, authoring 100 natural-language prompt suites to capture UI screenshots and telemetry on shared HPC infrastructure, identifying a 20% pass rate that drove improvements to model accuracy and prompt handling
- Prototyped LangChain agentic workflows connecting segmentation models to a surgeon-facing interface, validating feasibility for clinical deployment
- Delivered weekly technical presentations to biomedical experts, conducting EDA across BraTS medical imaging datasets to inform model selection and architecture recommendations for an agentic tumor segmentation system

Undergraduate Research Assistant

Jan 2026 - Jun 2026

Lijing Wang Lab | NJIT

- Designed controlled isolation experiments studying training instability and variance in LLM fine-tuning pipelines across Llama 3.2-3B, RoBERTa, and BERT on NJIT's Vulver A100 HPC cluster
- Diagnosed and resolved multiple sources of RNG inconsistency across data ordering, model initialization, and dropout in the training pipeline, preventing invalidation of 400+ planned runs by redesigning affected components with per-factor seed control

Deep Learning Forecasting Research Intern

Aug 2025 - Jan 2026

Grace Hopper AI Research @ NJIT

- Engineered modular Python preprocessing routines (NumPy, pandas) for cleaning, trimming, and interpolating 10+ multi-location epidemic datasets (~900 weekly observations each), creating consistent train/test splits and enabling reliable rolling-origin forecasting across 5 disease types
- Parallelized forecasting workflows with multi-threaded Python execution, reducing runtime by ~80% and enabling rapid model experimentation across diseases and hyperparameters without HPC resources
- Built project-wide logging utilities to automate metric collection and mid-run recovery, replacing manual aggregation and enabling reproducible experimentation at scale across datasets and model families

TECHNICAL PROJECTS

SHPE NJIT APP | Developer | GitHub Actions, Supabase

- Own CI/CD for the SHPE NJIT chapter mobile app, architecting automated test and deploy pipelines via GitHub Actions to support 200+ active members

AI-TUTOR | Lead Developer | Python, GraphRAG, Docker, Flask, BeautifulSoup

- Developed a pedagogical tutor mapping concept prerequisites via a knowledge graph built from a custom BFS web crawler and nano-graphrag extended with domain-specific entity extraction, grounding responses in graph-retrieved context to prevent hallucinations
- Implemented a rate-limited LLM wrapper with exponential backoff for reliable request handling, exposed via Flask backend and Streamlit UI on a Docker development container configured for selective service loading

EXTRACURRICULAR ACTIVITIES

Society of Hispanic Professional Engineers | NJIT

Jan 2024 - Present

Break Through Tech AI Fellow | Cornell Tech

Jun 2025 - Present

Google Software Engineering Fellow (Selected 2x) | G-SWEP

Jul 2025 - Sep 2025